

Article

Learning Low-Precision Structured Subnetworks Using Joint Layerwise Channel Pruning and Uniform Quantization

Xinyu Zhang ^{1,†}, Ian Colbert ^{2,†}  and Srinjoy Das ^{3,*} 
¹ Department of Computer Science, Rutgers University, New Brunswick, NJ 08901, USA

² Department of Electrical and Computer Engineering, University of California, San Diego, CA 92093, USA

³ School of Mathematical and Data Sciences, West Virginia University, Morgantown, WV 26506, USA

* Correspondence: srinjoy.das@mail.wvu.edu

† These authors contributed equally to this work.

Abstract: Pruning and quantization are core techniques used to reduce the inference costs of deep neural networks. Among the state-of-the-art pruning techniques, magnitude-based pruning algorithms have demonstrated consistent success in the reduction of both weight and feature map complexity. However, we find that existing measures of neuron (or channel) importance estimation used for such pruning procedures have at least one of two limitations: (1) failure to consider the interdependence between successive layers; and/or (2) performing the estimation in a parametric setting or by using distributional assumptions on the feature maps. In this work, we demonstrate that the importance rankings of the output neurons of a given layer strongly depend on the sparsity level of the preceding layer, and therefore, naïvely estimating neuron importance to drive magnitude-based pruning will lead to suboptimal performance. Informed by this observation, we propose a purely data-driven nonparametric, magnitude-based channel pruning strategy that works in a greedy manner based on the activations of the previous sparsified layer. We demonstrate that our proposed method works effectively in combination with statistics-based quantization techniques to generate low precision structured subnetworks that can be efficiently accelerated by hardware platforms such as GPUs and FPGAs. Using our proposed algorithms, we demonstrate increased performance per memory footprint over existing solutions across a range of discriminative and generative networks.

Keywords: channel pruning; layerwise pruning; quantization; joint pruning; quantization



Citation: Zhang, X.; Colbert, I.; Das, S. Learning Low-Precision Structured Subnetworks Using Joint Layerwise Channel Pruning and Uniform Quantization. *Appl. Sci.* **2022**, *12*, 7829. <https://doi.org/10.3390/app12157829>

Academic Editors: Deliang Fan and Zhezhi He

Received: 1 July 2022

Accepted: 23 July 2022

Published: 4 August 2022

Publisher's Note: MDPI stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Copyright: © 2022 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

The performance of deep neural networks (DNNs) has been shown to scale with the size of both the training dataset and model architecture [1]; however, the resources required to deploy larger networks for inference can be prohibitive as they often exceed the compute and storage budgets of resource-constrained platforms such as mobile or edge devices [2,3]. Therefore, as the usage of deep learning has proliferated in real-time applications with tight energy consumption budgets and low latency requirements, the field of research focused on reducing inference costs while maintaining model performance has rapidly expanded in recent years. Of the many techniques studied to accomplish this task, *pruning* and *quantization* are the most widely used and are often complementary to other approaches such as network distillation [4] and neural architecture search [5,6].

Pruning (i.e., the process of removing identified redundant elements from a neural network) and quantization (i.e., the processing of reducing their precision) are often considered to be independent problems [7,8]; however, recent work has begun to study the application of both in either a joint [2,6,9,10] or unified [11,12] setting. *Unified* algorithms typically use mixed precision quantization and integrate pruning by reducing the precision of an element (or a set of elements) to 0. On the other hand, *joint* algorithms combine separate optimization objectives for pruning and quantization under one learning framework, often